Fiscal Year:	FY 2017	**Task Last Updated:**	FY 11/11/2016
PI Name:	Long, Marshall Ph.D.		
Project Title:	Coflow Laminar Diffusion Flame		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	01/01/2012	**End Date:**	12/31/2016
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Continued by "Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment--NNX17AD49A" with same investigators Drs. Marshall Long and Mitchell Smooke, period of performance 3/1/2017-2/28/2020. NOTE: Extended to 12/31/2016 per D. Stocker/GRC and NSSC information (Ed., 1/14/2015) NOTE: change in Period of Performance to 1/1/2012-12/31/2014 per NSSC information (previous POP was 10/1/11-9/30/14)--Ed., 12/30/2011		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Smooke, Mitchell Ph.D. (Yale University)		
Grant/Contract No.:	NNX11AP43A		
Performance Goal No.:			
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Task Description:	Editor's Note: Successor agreement to "Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment (NNX07AB33A)," per D. Stocker, NASA Glenn Research Center. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth, and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place.
Task Progress:	Microgravity Experiments: SLICE and CLD Flame The Structure and Liftoff in Combustion Experiment (SLICE) was carried out on board the International Space Station (ISS) early in 2012. SLICE was a precursor to the Coflow Laminar Diffusion (CLD) Flame experiment, which is one of the experiments in the Advanced Combustion via Microgravity Experiments (ACME), slated to begin in 2017. As will be detailed in the following sections, we continued to build on techniques developed for SLICE in preparation for CLD Flame and ACME. Extension of High Dynamic Range (HDR) Imaging Techniques The HDR imaging concept, developed to analyze results from SLICE and described in last year's report, was extended to Rayleigh measurements in time-varying/turbulent flows. Two sequential laser pulses were used as illuminating sources and the Rayleigh-scattered light was imaged and recorded using the double frame option of an interline transfer CCD camera. Laser energies were such that the first image covered the majority of the detector dynamic range without reaching any saturation, while the second one was partially saturated in order to increase the pixel counts of the low signal level regions; the inter-pulse separation was chosen to be smaller than the characteristic time scales of the phenomena. The reconstructed HDR image showed increased signal-to-noise ratio (SNR) when compared to the unsaturated one, and equal or better SNR when compared to a conventionally averaged image. Additionally, an alternative image readout approach was demonstrated to improve the signal-to-noise ratio in temporally resolved laser-based imaging experiments. This latter method exploited the temporal decay characteristics of the phosphor screen of an image intensifier when coupled to an interline transfer CCD camera operated in double-frame mode. The light emitted by the phosphor screen was equally distributed and recorded over the two sequential frames of the detector so that an averaged image could be reconstructed. Again, its applicability to reactive flows was verified using planar Rayleigh scattering and tested with the acquisition of images of partially premixed methane/air flames. The comparison between conventional Rayleigh results and the averaged ones showed that the SNR of the latter was 30% higher than the former. This work was published in the journal Experiments in Fluids. CLD Flame Burner: Preliminary Characterization of the Flight and Ground Units The first iterations of the flight and engineering copies of the ACME CLD Flame hardware have undergone preliminary characterization. Hot wire anemometry measurements have been performed to assess the symmetry of the flows according to the ACME Integrated Science Requirements Document (for various flow conditions and at different downstream positions), and images of nitrogen-diluted coflow laminar diffusion methane flames were taken as a function of flow rates and fuel dilutions. The unsatisfactory flame symmetry that was observed at high lift-off demanded a revision of the burner coflow internal components (flow straightener and honeycomb). Several component configurations were manufactured and tested in-house at Yale, and the most satisfactory one is currently being replicated in the flight and engineering units at NASA Glenn. A replica of the ACME CLD Flame burner has also been manufactured at Yale so that parallel measurements can be taken with the SLICE detector and used to corroborate the results obtained with the ACME color data camera. Images of nitrogen-diluted methane and ethylene flames have been collected using the replica burner as a function of the predicted pressure increase inside the combustion integrated rack (from 1.0 to 1.3 atm, with 0.1 atm steps); the flame luminosity was then compared to the image exposure to derive a correlation between pressure and expected maximum camera counts for various flow conditions. ACME Detector: Flight Unit Spectral and Absolute Light Calibration The calibration of the signal from the flight version of the ACME detector to a source with a known intensity and spectral distribution is necessary for the determination of quantities such as soot volume fraction and CH* mole fraction. These calibrations were done off-site in conjunction with personnel at Zin Technologies. Two different approaches -- that span different temperature ranges -- were implemented, namely blackbody and thermocouple measurements. In both cases, images of a heated source (either a blackbody furnace or a thermocouple junction placed in a non-sooty premixed propane/air flame) were taken with the ACME setup. Color ratios were evaluated from the red/green/blue channels of the detector and compared with theoretical values obtained by employing Planck's law. When collected through a blue-green color filter (BG7), the comparison between the flight unit and the Yale detector showed a

discrepancy between the detectors' spectral responses that can be addressed by adjusting the transmissivity of the red and blue channels through the use of a multiplier. The results obtained without the color filter do not agree with the theoretical predictions and an investigation of the reason is currently underway.

Software Development and Documentation

As part of our NASA diagnostics development work, we have implemented techniques for quantitative measurements of soot temperature, soot volume fraction, thin-filament fiber temperature, and concentration of CH* in axisymmetric flames. These techniques require software for image acquisition and processing as well as careful characterization of the detection systems used. The bulk of this is done using image processing software developed in our labs over a period of more than thirty years. The “OMA” software undergoes continuous revision and extension and a version of the software for the Mac operating system has been available as open source since 2005 (http://oma-x.org). All of the ACME experiments will use the same detectors and many will be able to use the same processing methods that we have developed to handle the raw data. Therefore, the detector calibrations and latest version of the processing software are now available to all ACME investigators (and others as well through our open source web site). We have prepared a User’s Guide that outlines the basic processing steps for soot temperature and volume fraction determination. The user’s guide includes examples of raw data and the steps required to obtain quantitative measurements as well as macro scripts consisting of basic OMA commands, which will streamline the data processing. While the most advanced user interface and display capabilities are for the Mac operating system, versions for the Linux and Windows operating systems are available that implement all of the image processing algorithms and basic display capabilities.

Due to the relatively low pixel density of the ACME detector, interpolating algorithms (a.k.a. demosaicing algorithms) have been tested as a way to maintain the original spatial resolution when converting raw images into color ones. The “bilinear interpolation” and the “Malvar algorithm” have been implemented and added to OMA and their effectiveness tested against soot temperature measurements in a reference nitrogen-diluted ethylene flame. The quantitative information returned by the demosaiced images was seen to be more accurate that the one computed without any interpolation, albeit with an increase in the overall image noise.

Finally, a copy of the ACME software was installed on two computers here at Yale to closely simulate the way the ACME images will be taken on board the ISS and to test the ability of the detector to be controlled remotely. Moreover, OMA is now capable of decoding the .hobj image format that will be used to save the raw microgravity images.

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Bibliography Type:	Description: (Last Updated: 01/12/2026)
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Articles in Peer-reviewed Journals	Giassi D, Cao S, Bennett BAV, Stocker DP, Takahashi F, Smooke MD, Long MB. "Analysis of CH* concentration and flame heat release rate in laminar coflow diffusion flames under microgravity and normal gravity." Combustion and Flame. 2016 May;167:198-206. http://dx.doi.org/10.1016/j.combustflame.2016.02.012 , May-2016
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